LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need for Voltage regulation, types of voltage Regulators - Carbon pile voltage regulator, solid voltage regulators, paralleling and load sharing
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need for Voltage regulation, types of voltage Regulators - Carbon pile voltage regulator, solid voltage regulators, paralleling and load sharing.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the A/C power systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Carbon pile regulator principle; Safety: High voltage precautions; Supervision: Instructor monitors connections
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft power system?
(b) Why is stable voltage critical for avionics operation?
(c) What happens if generator output voltage fluctuates significantly?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need for Voltage regulation • Carbon pile voltage regulator • Solid state voltage regulators • Paralleling generators • Load sharing principles	L & Ds	PPT, IB	15 min
(b)	• Carbon pile regulator operation • Solid state regulator advantages • Paralleling circuit configuration • Load sharing mechanisms • Regulator failure modes	L & Ds	PPT, PDF	20 min
(c)	• Voltage regulation standards • Maintenance checks • Troubleshooting procedures • Safety protocols • System integration	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the working principle of a carbon pile voltage regulator.
(b) Describe the benefits of solid state regulators over carbon pile types.
(c) How is load sharing achieved when paralleling generators?

12. Home Assignment and Follow Up	Prepare a brief note on the battery connection points in the aircraft to be discussed in the next lesson on Batteries.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Batteries- functions, different types of batteries used in A/C, Location of batteries in A/C battery connection, typical battery systems, on board battery charging and warning system and memory effect on NI- Cd batteries. Practical in battery shop. Safety precautions to be observed in battery shop.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min